

● MEDIUM

● EXPRESSION OF IDEAS

Expression of Ideas

08

10 questions · ~15 min

Q1

EXPRESSION OF IDEAS

In 2021, a model developed by astrophysicist Catherine Zucker and her research team revealed that the same supernovas responsible for the creation and ongoing expansion of the Local Bubble—a 14-million-year-old cavity in the Milky Way—are likely responsible for the formation of new stars. _____blank this model detailed how the bubble’s expansion trapped interstellar clouds of gas and dust that became stars upon their eventual collapse.

Which choice completes the text with the most logical transition?

- A. Hence,
- B. However,
- C. Admittedly,
- D. Specifically,

While researching a topic, a student has taken the following notes:

- John Carver was one of the 41 signatories of the Mayflower Compact.
- The Mayflower Compact was a legal agreement among the pilgrims that immigrated to Plymouth Colony.
- It was created in 1620 to establish a common government.
- It states that the pilgrims who signed it wanted to “plant the first colony in the northern parts of Virginia” under King James.
- Carver became the first governor of Plymouth Colony.

The student wants to specify the reason the Mayflower Compact was created. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Stating that its signatories wanted to “plant the first colony in the northern parts of Virginia,” the Mayflower Compact was a legal agreement among the pilgrims that immigrated to Plymouth Colony.
- B. Created in 1620, the Mayflower Compact states that the pilgrims wanted to “plant the first colony in the northern parts of Virginia.”
- C. The Mayflower Compact was created to establish a common government among the pilgrims that immigrated to Plymouth Colony.
- D. The Mayflower Compact had 41 signatories, including John Carver, the first governor of Plymouth Colony.

In 2019, researcher Patricia Jurado Gonzalez and food historian Nawal Nasrallah prepared a stew from a 4,000-year-old recipe found on a Mesopotamian clay tablet. When they tasted the dish, known as *pašrūtum* (“unwinding”), they found that it had a mild taste and inspired a sense of calm. _____ blank the researchers, knowing that dishes were sometimes named after their intended effects, theorized that the dish’s name, “unwinding,” referred to its function: to help ancient diners relax.

Which choice completes the text with the most logical transition?

- A. Therefore,
- B. Alternately,
- C. Nevertheless,
- D. Likewise,

While researching a topic, a student has taken the following notes:

- Etel Adnan was a Lebanese American poet and artist known for making many leporellos.
- A leporello is an artist's book that is folded accordion style.
- When the book is expanded, the artist's work is revealed, and its zigzag shape allows it to stand on its own.
- Her leporello *December from My Window* (1993) features a panoramic landscape.
- It is painted using ink and watercolor.

The student wants to describe Adnan's *December from My Window* to an audience already familiar with leporellos. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Featuring a panoramic landscape, the 1993 work is one of Adnan's many leporellos, which are accordion-style folded books that when expanded reveal the artist's work.
- B. When expanded, Adnan's 1993 leporello *December from My Window* reveals a panoramic landscape painted in ink and watercolor.
- C. Known for making many other accordion-style folded books called leporellos, Adnan created *December from My Window* in 1993.
- D. A leporello, such as Adnan's *December from My Window*, is folded accordion style, and due to its zigzag shape it is able to stand on its own when fully expanded.

In the 1880s, inventor Lewis Latimer improved upon Thomas Edison's design for the electric light bulb. _____blank Latimer made the light bulb more durable by placing cardboard around its carbon filament. With this innovation, Latimer became the first Black inventor to contribute to the electrification of the world.

Which choice completes the text with the most logical transition?

- A. Soon,
- B. Regardless,
- C. However,
- D. Specifically,

While researching a topic, a student has taken the following notes:

- Pterosaurs were flying reptiles that existed millions of years ago.
- In a 2021 study, Anusuya Chinsamy-Turan analyzed fragments of pterosaur jawbones located in the Sahara Desert.
- She was initially unsure if the bones belonged to juvenile or adult pterosaurs.
- She used advanced microscope techniques to determine that the bones had few growth lines relative to the bones of fully grown pterosaurs.
- She concluded that the bones belonged to juveniles.

The student wants to present the study and its findings. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. In 2021, Chinsamy-Turan studied pterosaur jawbones and was initially unsure if the bones belonged to juveniles or adults.
- B. Pterosaur jawbones located in the Sahara Desert were the focus of a 2021 study.
- C. In a 2021 study, Chinsamy-Turan used advanced microscope techniques to analyze the jawbones of pterosaurs, flying reptiles that existed millions of years ago.
- D. In a 2021 study, Chinsamy-Turan determined that pterosaur jawbones located in the Sahara Desert had few growth lines relative to the bones of fully grown pterosaurs and thus belonged to juveniles.

The chemical trimethylamine N-oxide not only gives fish their fishy smell but also protects them from crushing hydrostatic pressure in deep waters. Trimethylamine N-oxide strengthens the bonds between water molecules in a fish's body. _____ blank these water molecules maintain their linked structure at extreme depths, thus preventing pressure-related damage.

Which choice completes the text with the most logical transition?

- A. Nevertheless,
- B. As a result,
- C. However,
- D. For instance,

- In 2017, a research team led by Mary Caswell Stoddard determined the average lengths of eggs produced by various bird species.
- *Gygis alba* is a species of bird in the order Charadriiformes.
- *Gygis alba* eggs had an average length of 4.46 cm.
- *Gavia stellata* is a species of bird in the order Gaviiformes.
- *Gavia stellata* eggs had an average length of 7.22 cm.

Which choice most effectively uses information from the given sentences to emphasize a difference between the eggs of the two species?

- A. A 2017 study compared the lengths of eggs produced by an array of different bird species, such as *Gygis alba* and *Gavia stellata*.
- B. A 2017 study found that *Gygis alba* eggs had an average length of 4.46 cm, whereas *Gavia stellata* eggs were longer, with an average length of 7.22 cm.
- C. The bird species *Gygis alba*, which belongs to the order Charadriiformes, and *Gavia stellata*, of the order Gaviiformes, were included in a 2017 study that compared the average lengths of their eggs.
- D. Mary Caswell Stoddard led a research study that determined the average lengths of eggs, including those of *Gygis alba* birds (4.46 cm) and *Gavia stellata* birds (7.22 cm).

In 1933, the Twentieth Amendment to the US Constitution was ratified. The amendment mandates that presidential inaugurations be held on January 20, approximately ten weeks after the November election. _____ blank this amendment requires newly elected US senators and representatives to be sworn into their respective offices on January 3.

Which choice completes the text with the most logical transition?

- A. Instead,
- B. For instance,
- C. Specifically,
- D. In addition,

While researching a topic, a student has taken the following notes:

- In 2022, University of Miami researchers discovered brine pools in the Gulf of Aqaba.
- A brine pool is an underwater lake that sits on the ocean floor.
- The water in brine pools is three to eight times saltier than the surrounding ocean.
- The extreme saltiness of this water makes it toxic to most sea life.
- Some forms of bacteria are able to survive in brine pools.

The student wants to explain why brine pools are toxic to most sea life. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Though brine pools are toxic to most sea life, some bacteria can survive there.
- B. The water in brine pools is toxic to most sea life because it is three to eight times saltier than the surrounding ocean.
- C. The brine pools in the Gulf of Aqaba are toxic to most sea life and were discovered by researchers in 2022.
- D. Brine pools are salty underwater lakes that sit on the ocean floor.